



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LAB 18 INSTRUCTIONS

For

Produce an evidence-led incident report

Course Code: C1123

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.1

Lab 18 — Produce an evidence-led incident report

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 18 — Ten Troubleshooting Steps and Reporting

Learning outcome: LO5 — Report observed facts, hypotheses and next actions with defensible evidence.

Time: about 75 minutes

Version: v5.1 · 5 October 2026

Scenario

Ticket BR-104 combines several user complaints. You produce a reproducible incident report that separates observed facts from hypotheses.

Goal

Begin with scope and a baseline, then focus with endpoints, filters, time and graphs.

What you will produce

An evidence-led incident report.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete
assets/incident-ticket.md	Incident ticket BR-104
assets/report-template.md	Report template
assets/ten-step-checklist.md	Ten-step review checklist

Lab at a glance

1. Read the incident ticket
2. Collect evidence for each symptom
3. Write the report from the template
4. Peer-review with the ten-step checklist

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap`. Read `assets/incident-ticket.md`. Use Statistics > Conversations, filters and Expert Information to identify the failing /fault request.

3. Record evidence for the DNS delay, slow HTTP response, retransmission, zero window and port 81 refusal. Use exact frames and elapsed times rather than a claim that every symptom has one cause.

4. Complete `assets/report-template.md` and save as `outputs/incident-report.md`. Include scope, timeline, observations, competing explanations, next capture placement and a client-safe recommendation.

5. Use `assets/ten-step-checklist.md` to review the report. Re-run `scripts/verify.py`. Exchange reports with another learner and check that every conclusion can be traced to a packet or is explicitly a hypothesis.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

http.response.code == 500					
No.	Time	Source	Destination	Protocol	Info
50	2.270000	192.0.2.20	192.0.2.10	HTTP	HTTP/1.1 500 Internal Server Error (text/plain)
branch-office.pcap		Displayed: 1			

Lab 18 expected evidence

Test it

The supplied `http.response.code == 500` expression matches 1 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.

- TLS remains opaque: select the matching lab-tls.keys file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -q -z conv,tcp
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

Write a second incident report from a capture in the Wireshark sample captures (wiki.wireshark.org/SampleCaptures), using the same template and checklist.

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.